Children and adults' inferences about social groups in response to sampling skew

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Social structures shape
who we see and don’t see.

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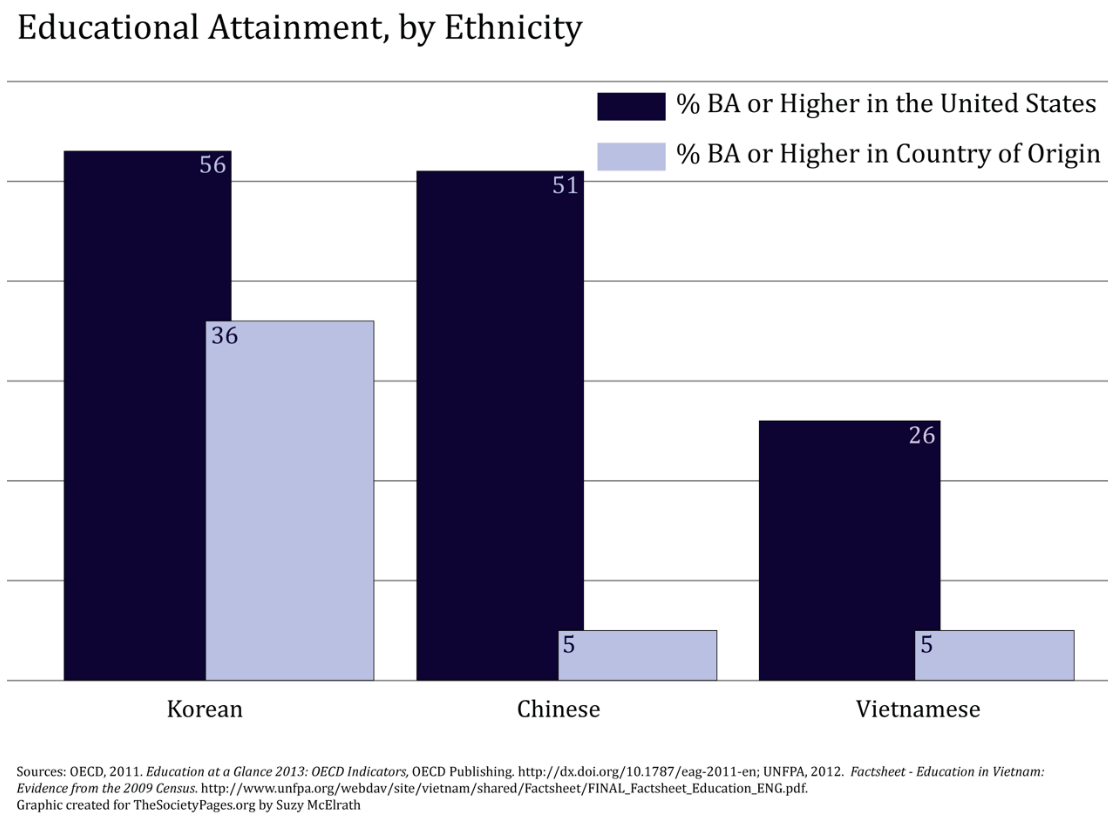
Chinese Americans are, on average, more **highly educated** and **high-achieving**, than counterparts in China.

due to **hyperselective US immigration policies** (e.g., 1965 Immigration & Nationality Act)



(plot made by sociologists, please excuse the axes)

Lee & Zhou (2015)

Social structures shape who we see and don’t see.

Chinese Americans are, on average, more **highly educated** and **high-achieving**, than counterparts in China.
(due to selective immigration policies)

↓

what is seen

American children tend to see disproportionately **highly-educated**, **high-achieving** Chinese people.

Lee & Zhou (2015)

Exposure to skewed samples might provide a rational basis for stereotypes.

Chinese Americans are, on average, more **highly educated** and **high-achieving**, than counterparts in China.
(due to selective immigration policies)

what is seen

American children tend to see disproportionately **highly-educated, high-achieving** Chinese people.

Problematic effects

- glosses over variation within the group
- Asians who need support may not get it
- harmful to Asian peoples' mental health
- ignores anti-Asian racism
- implies other racial minorities are lacking in comparison

model minority stereotype

belief that Chinese people (and, by extension, Asian people) are (inherently) intelligent & high-achieving

Exposure to skewed samples might provide a rational basis for stereotypes.

Surveillance cameras are disproportionately placed in Black neighborhoods.

what is seen

Footage of anti-Asian hate crimes disproportionately show Black perpetrators.

Black criminality stereotype

belief that Black people are (inherently) more likely to commit crimes (e.g., anti-Asian hate crimes)

Problematic effects

- more likely to be targeted by police violence
- greater presumption of guilt
- more severe sentencing
- more likely to be incarcerated
- …

Generalizing from skewed samples of social groups could lead to stereotypes.

Skewed samples as a potential basis for stereotypes

statistical inference

assuming the sample is randomly selected, generalize directly from sample to population

population

actual social group

sample

observe group members (selected for X) are X

population inference

infer that social group in general is X

sampling process

structural processes (e.g., immigration policies, segregation, media biases) skew who is seen and who is not seen

Skewed samples as a potential basis for stereotypes

~~statistical inference~~
~~assuming the sample is randomly selected,~~
~~generalize directly from sample to population~~

population

actual social group

sampling process

structural processes (e.g., immigration policies, segregation, media biases)
skew who is seen and who is not seen

sample

observe group members (selected for X) are X

population inference

infer that social group in general is **actually not as** X as observed

Why adjusting against skewed social samples might be early-emerging

Why adjusting against skewed social samples might be late-emerging

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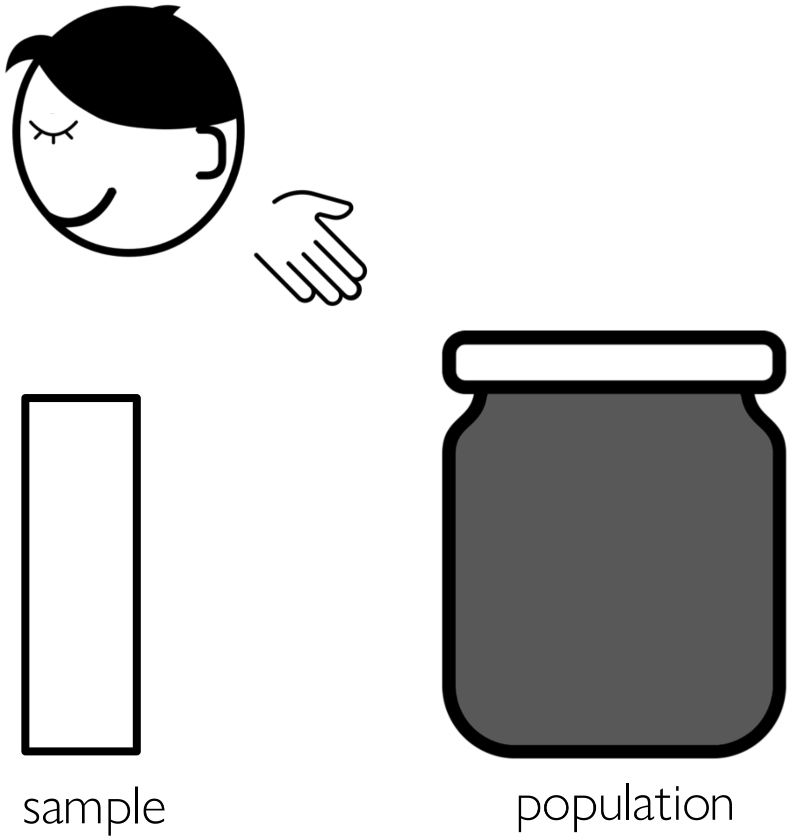
Even infants can recognize the statistics of samples!

Why adjusting against skewed social samples might be early-emerging



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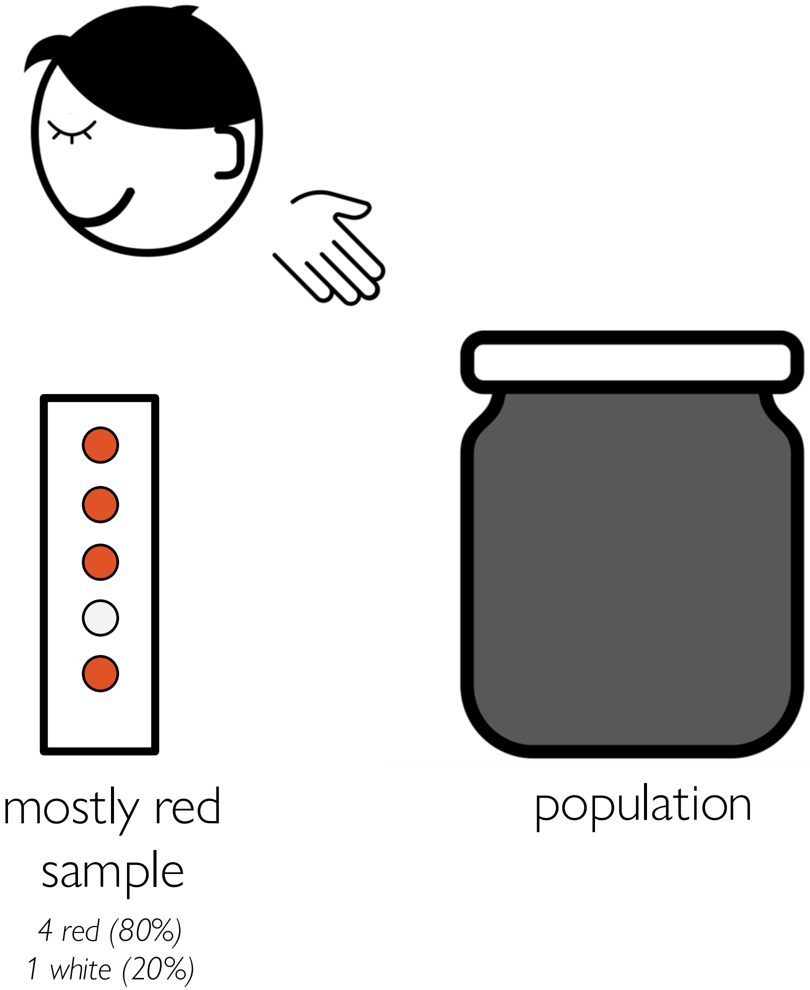
- generalize from a random sample to a population (marbles in a jar)

Xu & Garcia (2008)

Why adjusting against skewed social samples might be early-emerging



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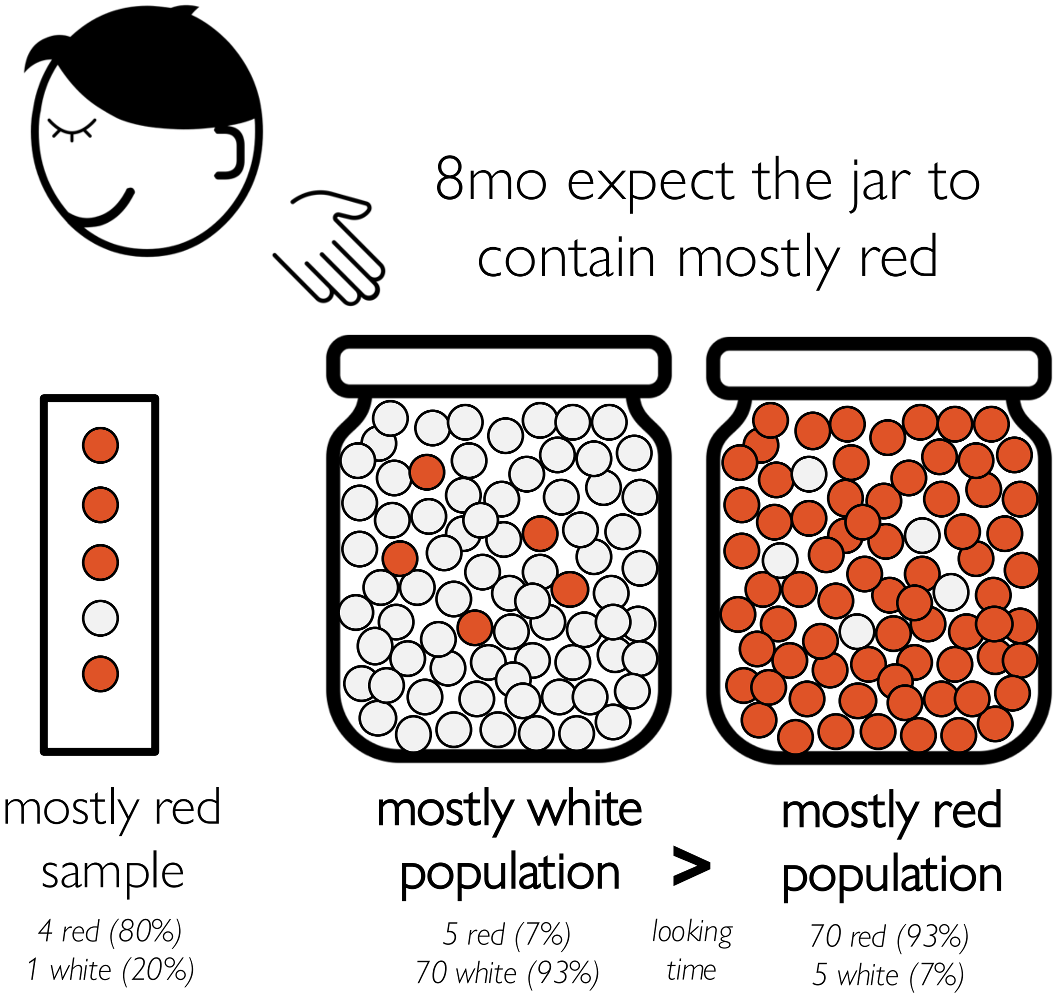

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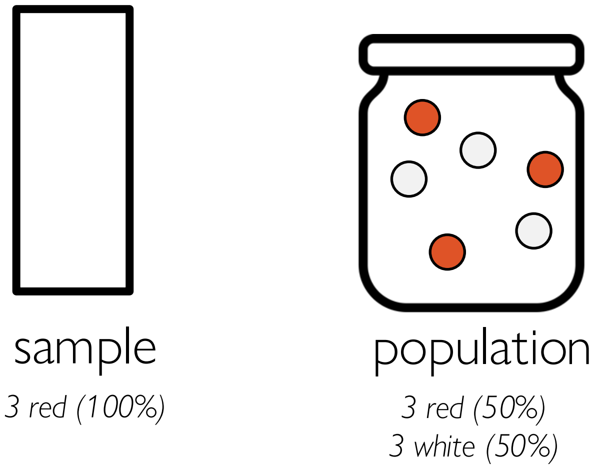

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- **generalize** from a **random sample** to a population (marbles in a jar)
- and **suspend inference** when a sample is **skewed** (by an agent's preferences)

Xu & Denison (2009)

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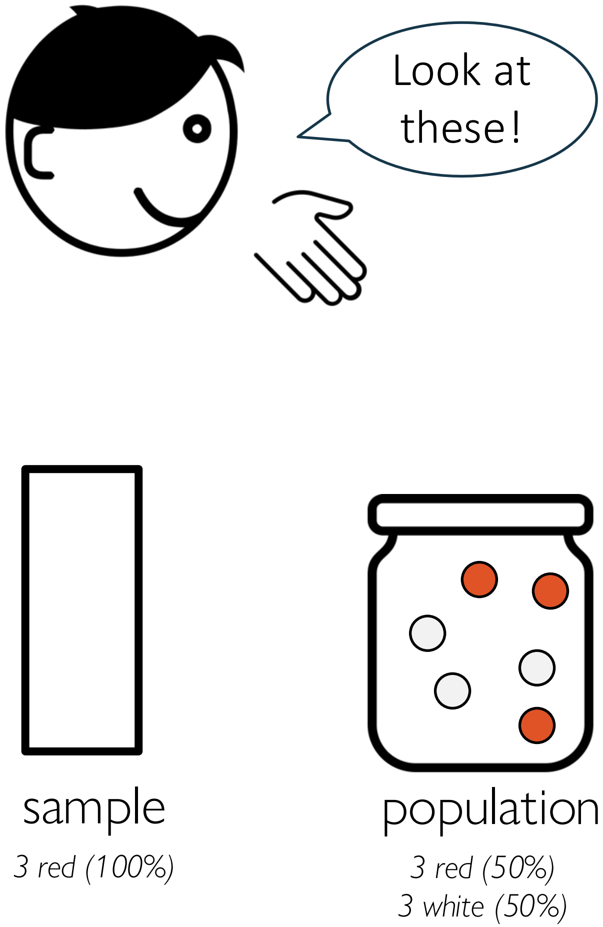

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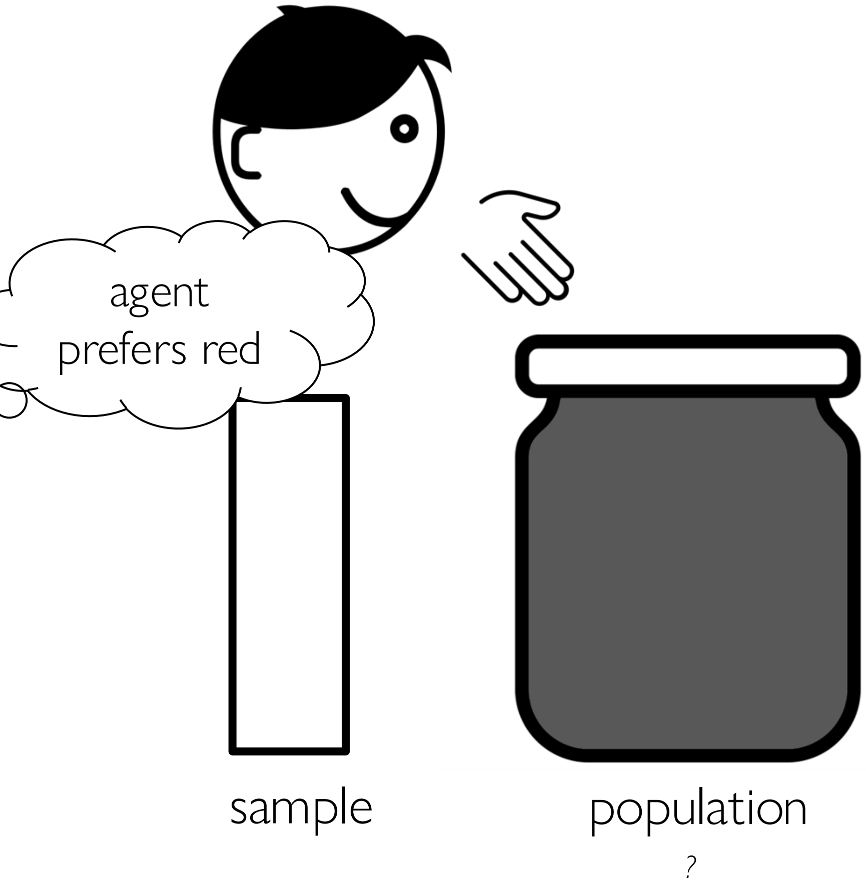

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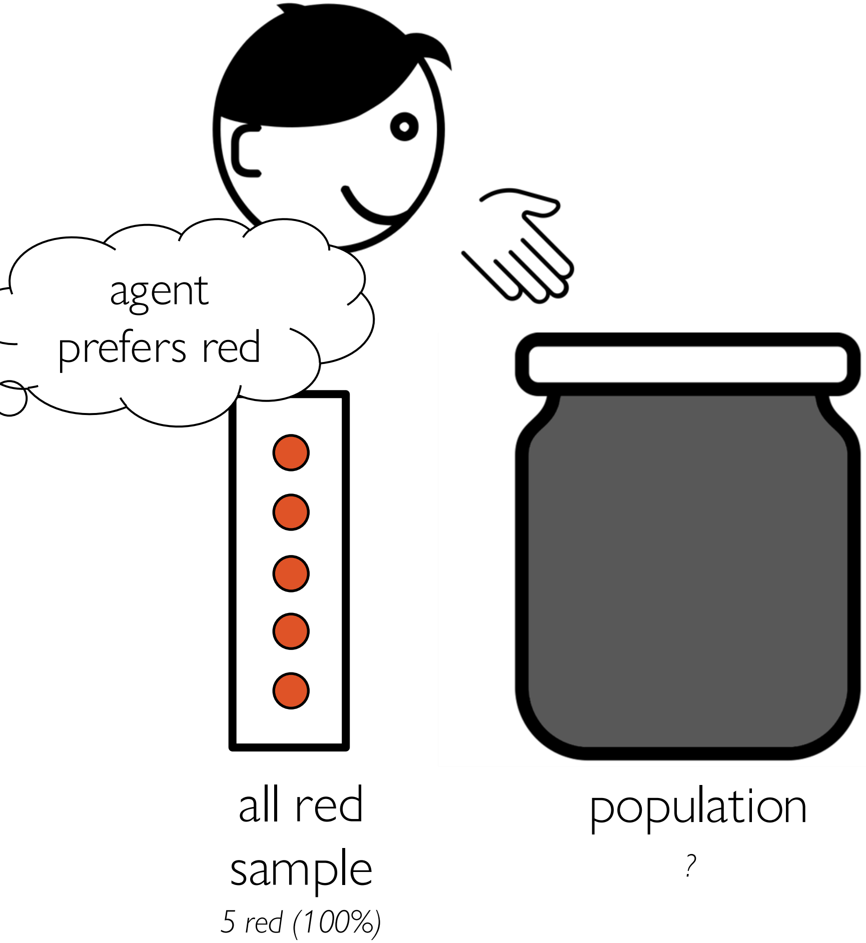

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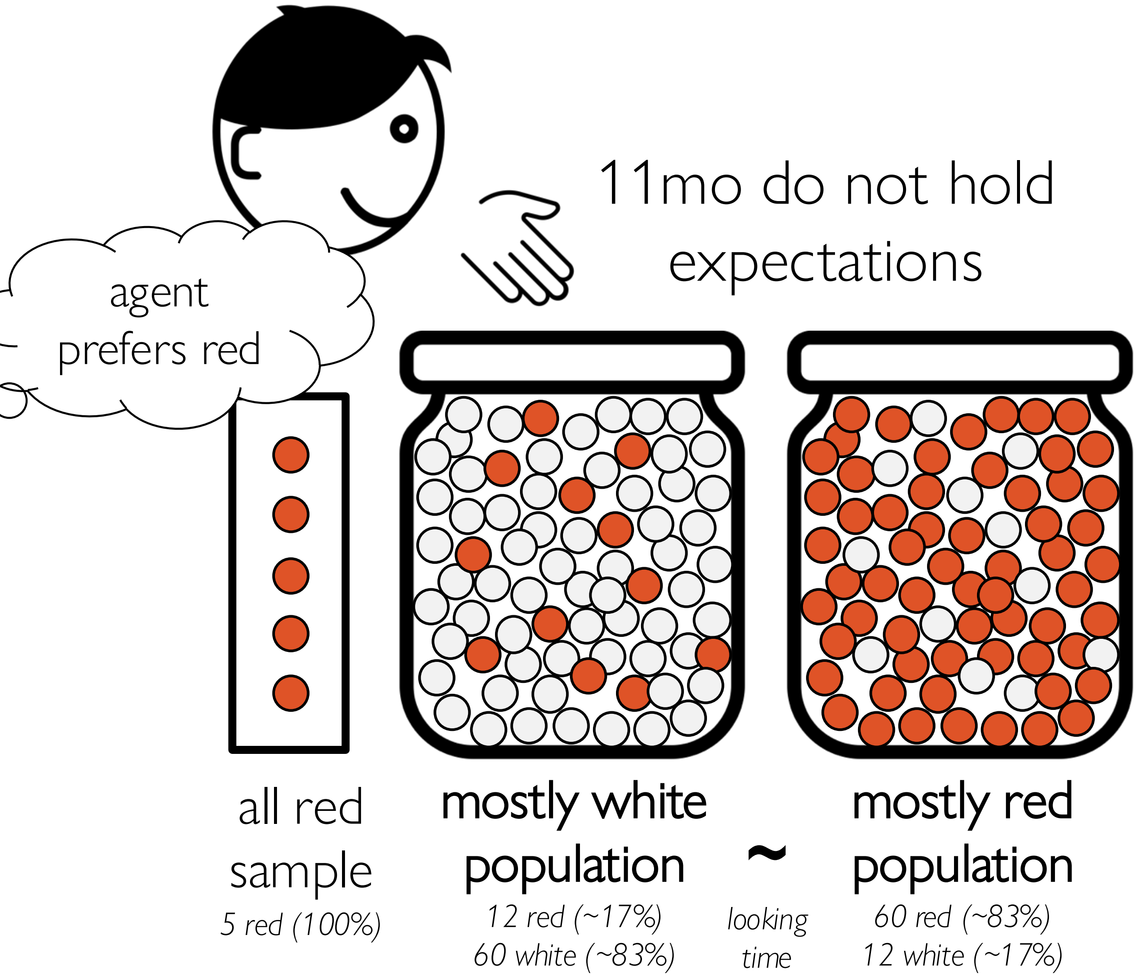

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- and **suspend inference** when a sample is **skewed** (by an agent's preferences)

Xu & Denison (2009)

Why adjusting against skewed social samples might be early-emerging

even infants can recognize skewed samples!

Why adjusting against skewed social samples might be late-emerging

Why adjusting against skewed social samples might be late-emerging

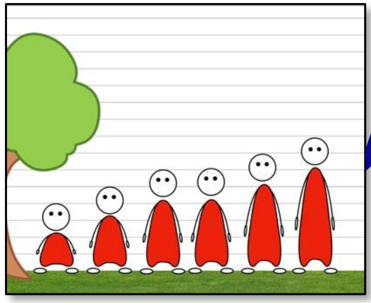
- Earlier tasks show that infants can **suspend** inference (in response to deterministic skew), but the ability to **adjust** inferences (against probabilistic skew) may be more challenging.
- Skew in the social domain is often caused by an **external structure**, rather than the intentionality of a single agent.
  - **Structural reasoning** is challenging before ages 5-7. (Vasilyeva et al., 2018; Peretz-Lange et al., 2021)

Can children & adults **adjust** their inferences against skewed social samples?

Study 1

adjusting social group inference from a skewed sample

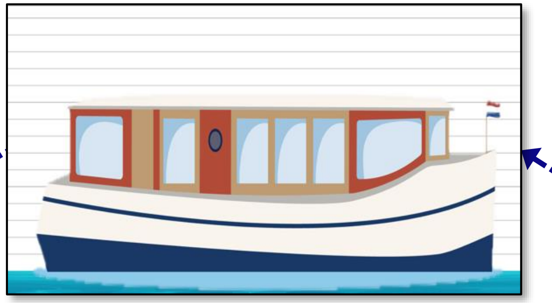
sample



observe Zarpies who come from Zarpie island by boat

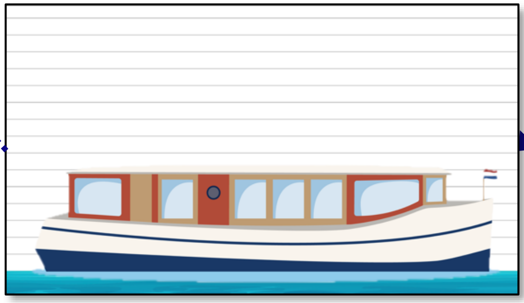
sampling process

not skewed



tall boat allowed Zarpies to board easily

skewed



short boat mostly restricted boarding to short Zarpies

population inference

infer height of Zarpies on Zarpie island

1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

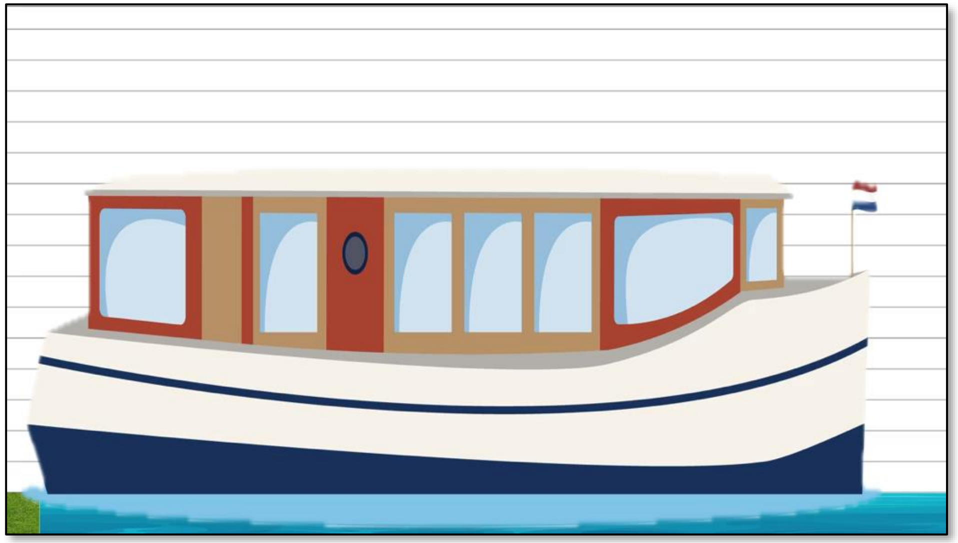
Study 1

adjusting social group inference from a skewed sample

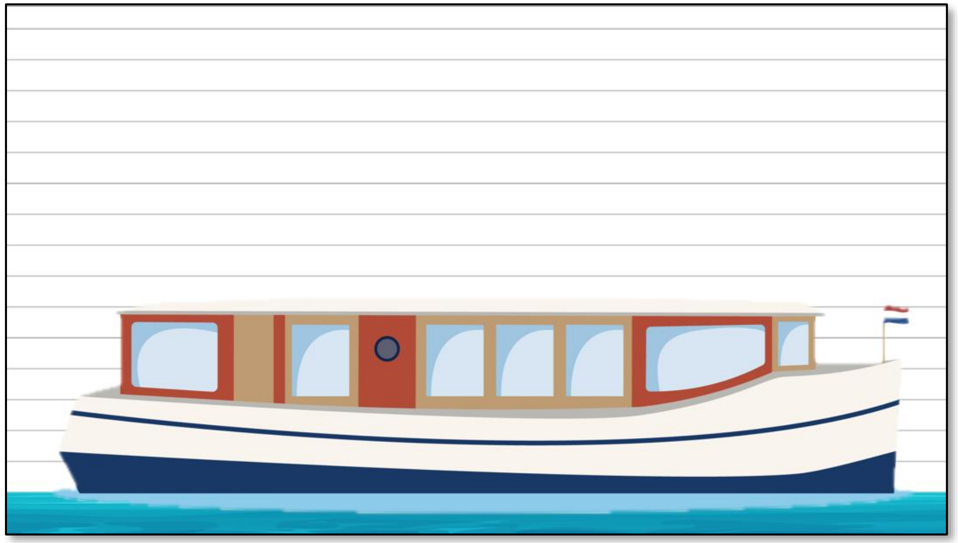
warmup | **training** | sampling process | sample observation | inference

learn that the boat height is incidental (uninformative to the height of its passengers)

not skewed



skewed



"When the boat builders were building the boat, they started building the boat from the bottom, but ran out of the special wood they needed for the boat! So the boat ended up being this tall. It might be hard for anyone who is taller than the boat to get on the boat."

1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

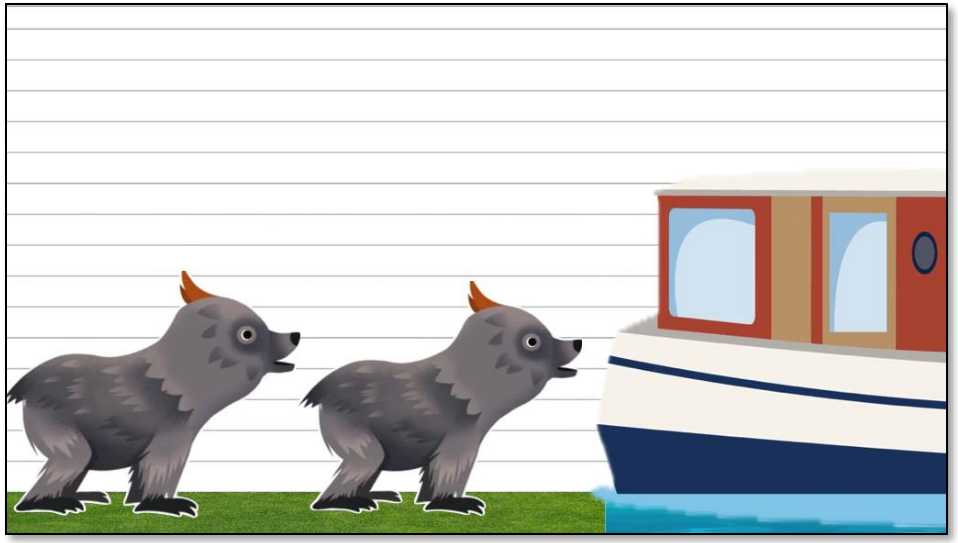
Study 1

adjusting social group inference from a skewed sample

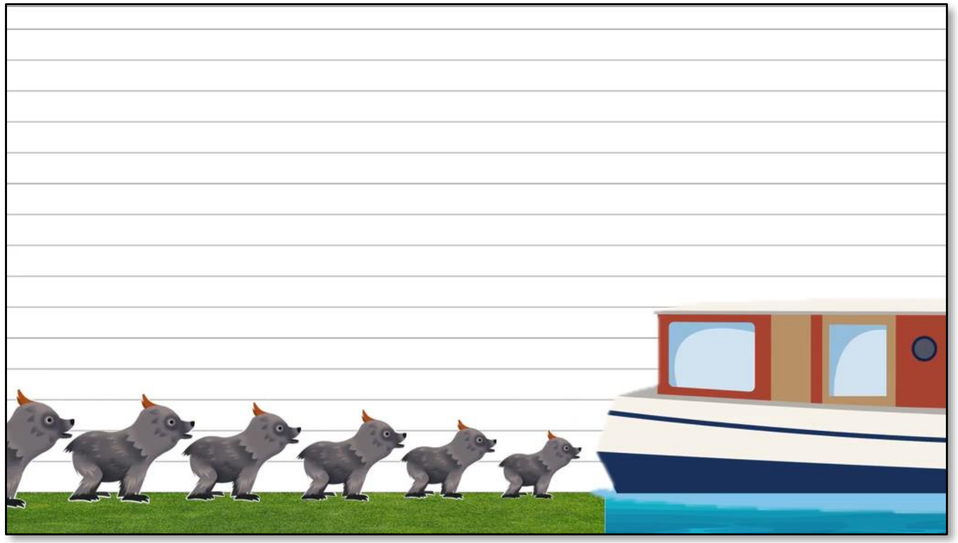
warmup	training	sampling process	sample observation	inference

learn that the boat probabilistically excludes anything taller

not skewed



skewed



20 fictional animals attempt to board the boat:

- Of the 10 animals shorter than the boat, all 10 (100%) make it on
- Of the 10 animals taller than the boat, only 1 (10%) makes it on (by scrunching)

1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

Study 1

adjusting social group inference from a skewed sample

warmup	training	sampling process	sample observation	inference

learn how Zarpies on Zarpie island are sampled

"Now, let's learn about Zarpies! Zarpies are people who live on a far-away island. See all these Zarpies on Zarpie island? Now, the grownup Zarpies' names were put into a hat, and some of their names were drawn out to try to visit us. Those Zarpies are going to try to board the boat that we saw before."

1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

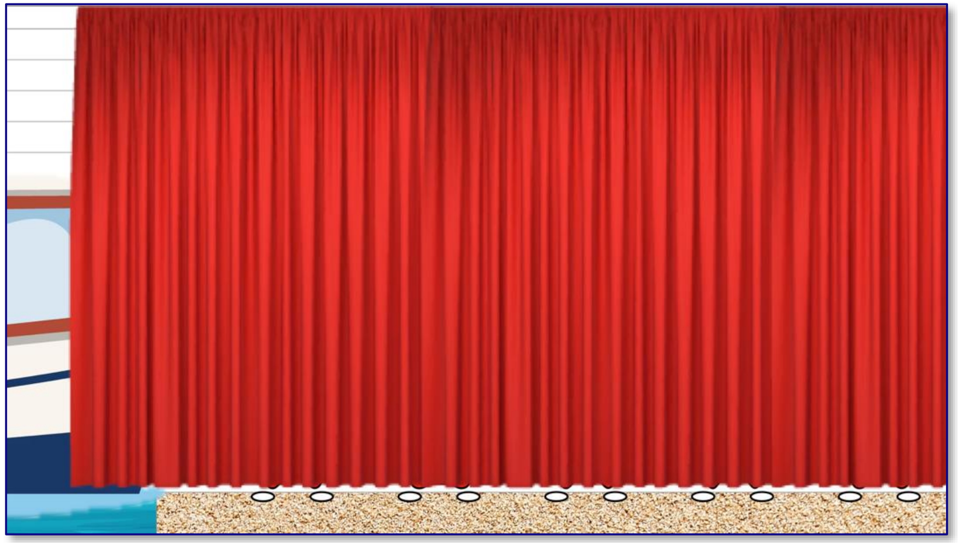
Study 1

adjusting social group inference from a skewed sample

warmup | training | **sampling process** | sample observation | inference

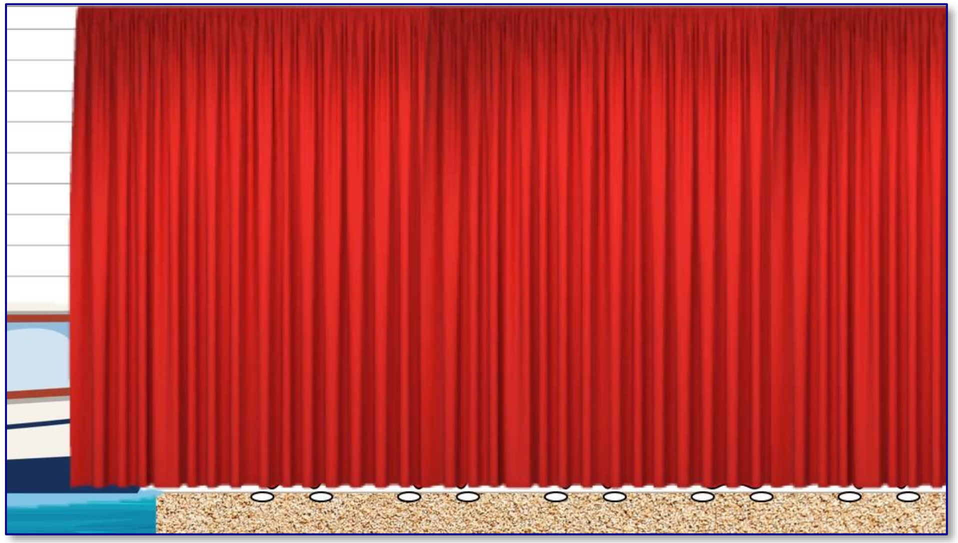
learn how Zarpies on Zarpie island are sampled

not skewed



tall boat allows Zarpies to board easily
6 out of **6** (100%) Zarpies board
(0 by scrunching)

skewed



short boat filters out many Zarpies
6 out of **20** (30%) Zarpies board
(2 by scrunching)

1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

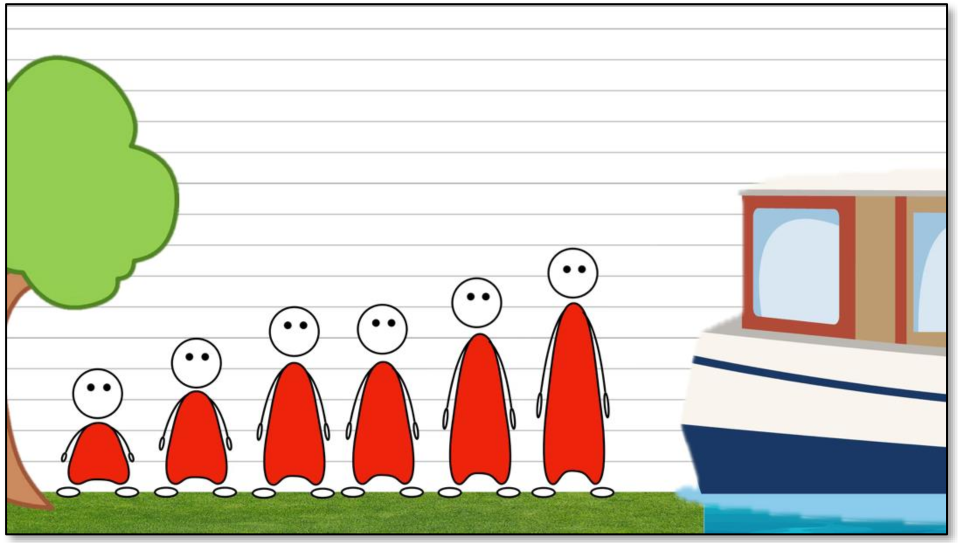
Study 1

social group inference from a skewed sample

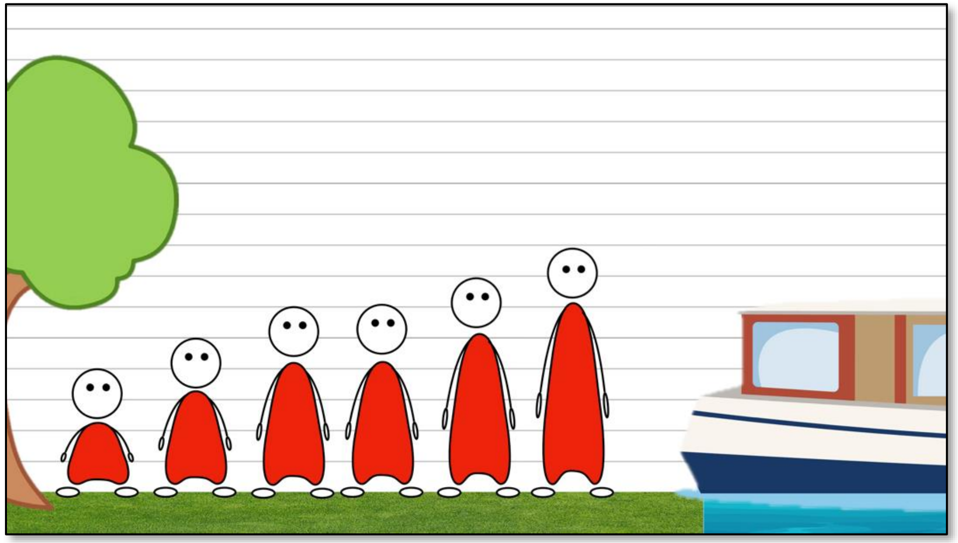
warmup | training | sampling process | **sample observation** | inference

observe a sample of Zarpies (the Zarpies from Zarpie island who successfully boarded the boat)

not skewed



skewed



observed Zarpies are identical in height across conditions, although the result of different sampling processes

1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

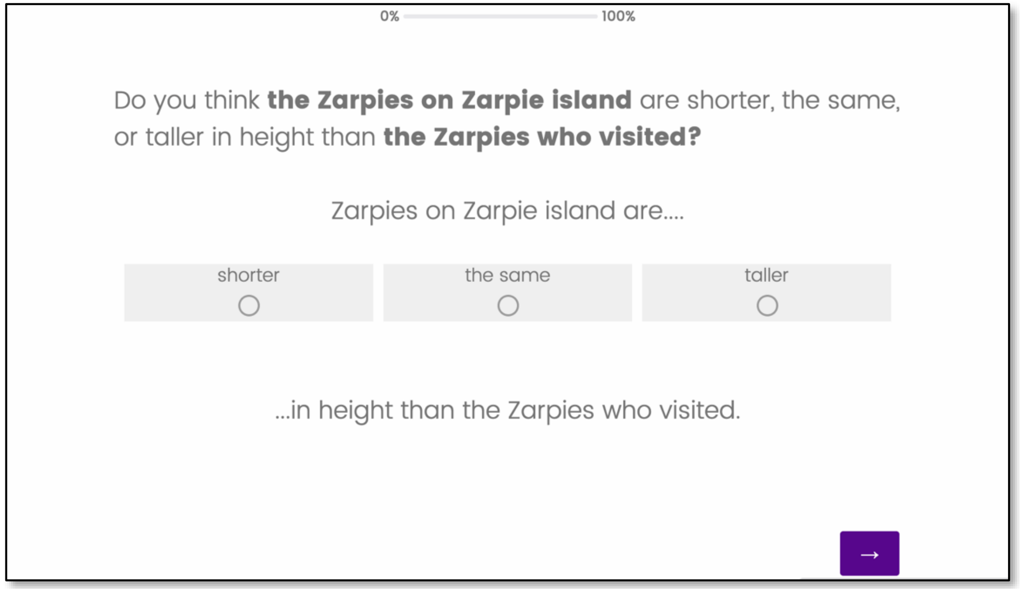
Study 1

social group inference from a skewed sample

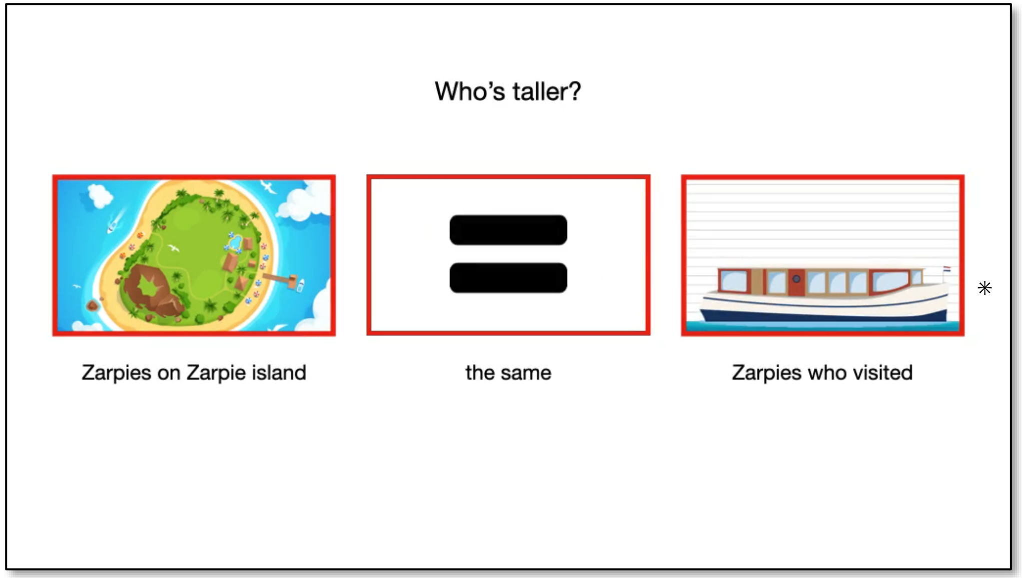
warmup | training | sampling process | sample observation | **inference**

infer height of Zarpies on Zarpie island (population),
as compared to the Zarpies who visited (observed sample)

adults



4-8yo



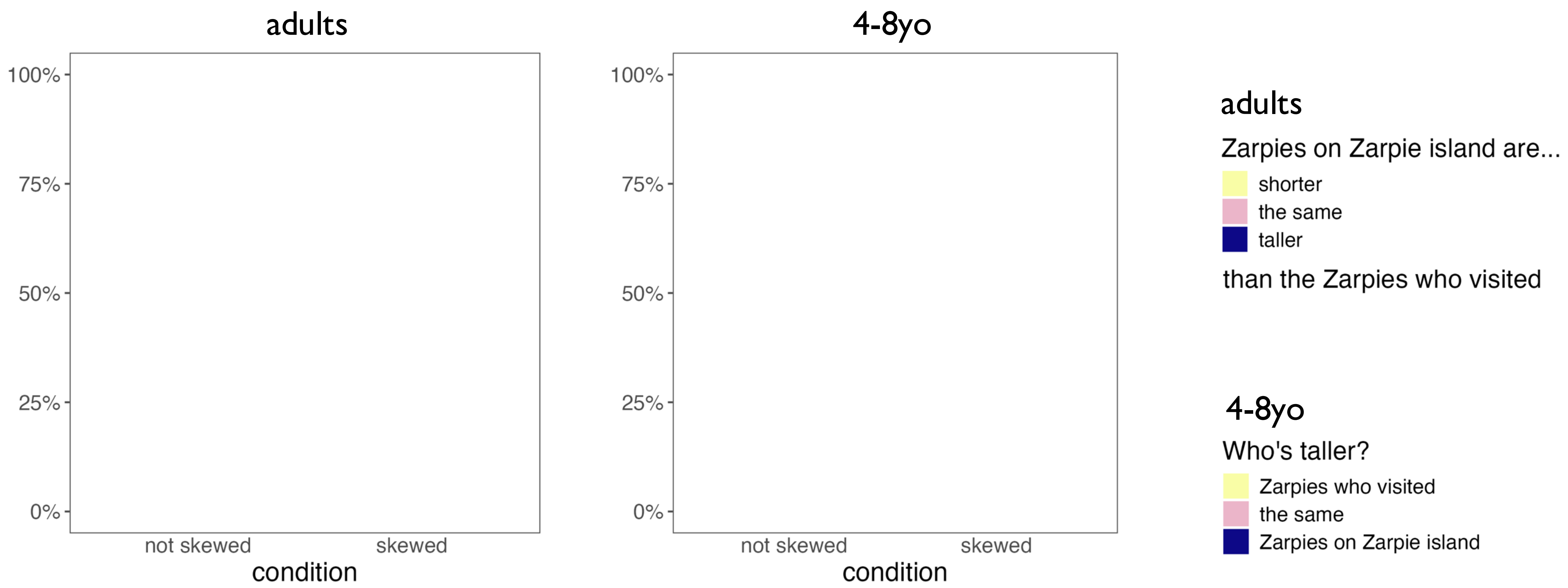
1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

*skewed condition shown; image of boat varied across conditions
for children, order of options counterbalanced across participants

Study 1

if adults and children can adjust inferences against skewed sampling…

adults and children will be more likely to infer that the social group is taller when the sample was skewed towards shorter individuals, than when the sample was not skewed.



1a: **adults** (n=100/condition) on Prolific
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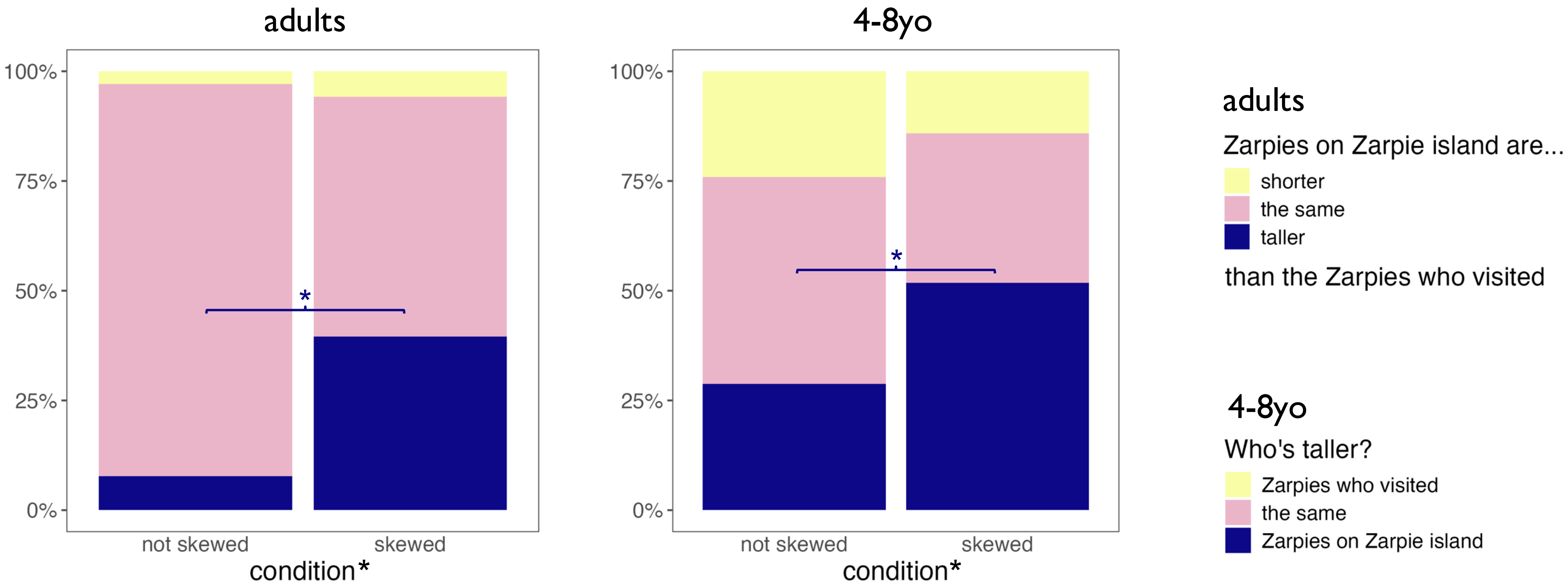
Study 1

evidence for adjustment against skewed sampling

adults and children were more likely to infer that the social group was taller when the sample was skewed towards shorter individuals, than when the sample was not skewed.



1a: **adults** (n=100/condition) on Prolific
1b: **4-8yo** (n=85/condition) on PANDA

all responses ~ condition: Fisher's exact: p=.008
"Zarpies on Zarpie island" ~ condition: z=3.05, p=.002

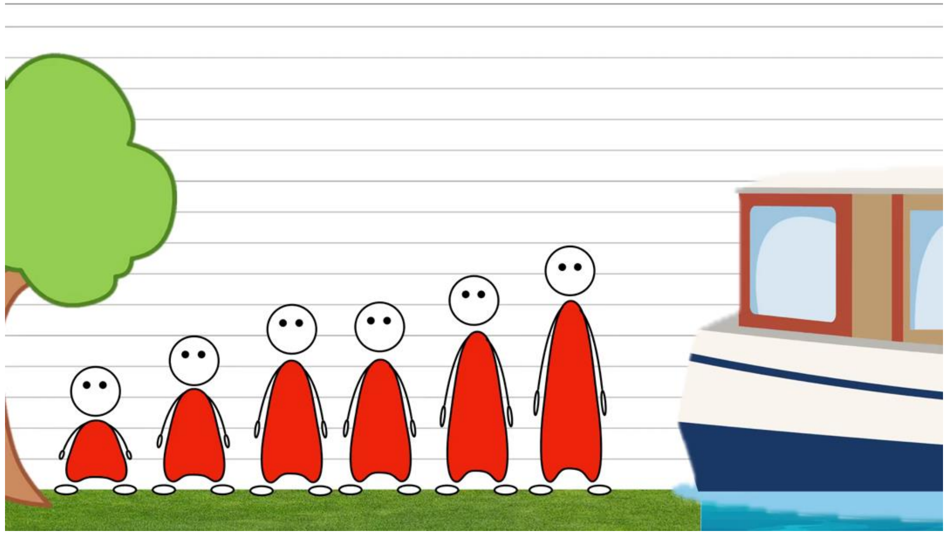
Study 1

adjusting social group inference from a skewed sample

alternative explanation: perceptual salience

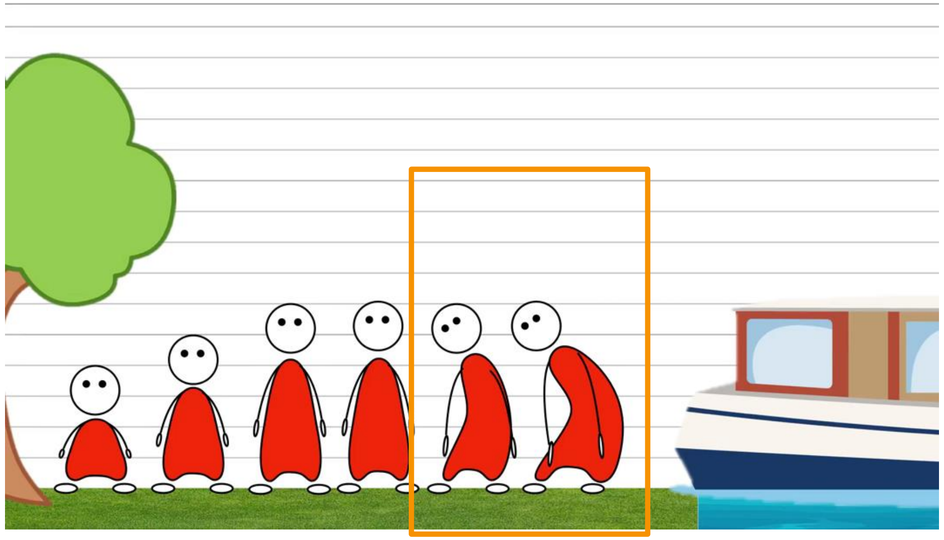
adults and children ignored sampling processes, and **generalized directly from the sample, weighting the taller Zarpies more in the skewed condition**, since scrunching makes them salient

not skewed



each Zarpie in sample weighted equally

skewed



tallest two Zarpies in sample weighted more?

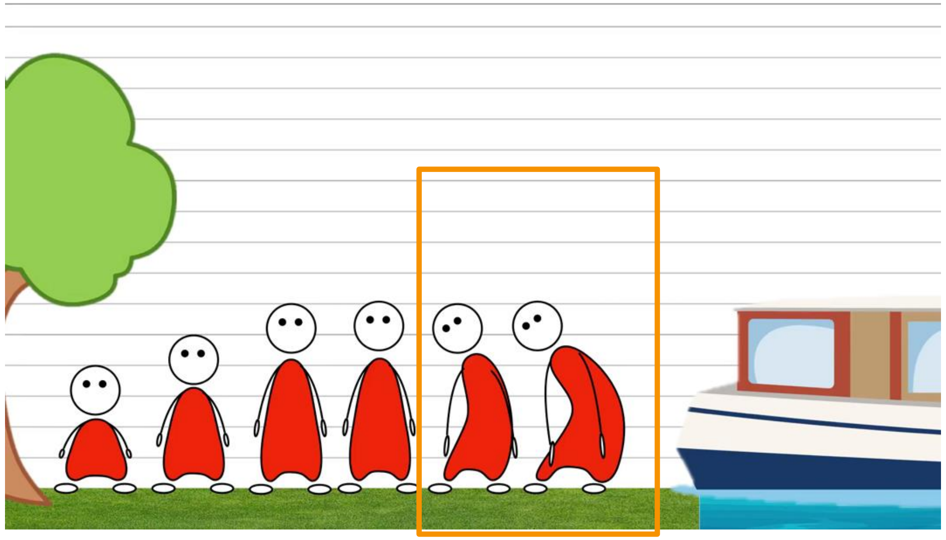
Study 2

addressing an alternative perceptual explanation

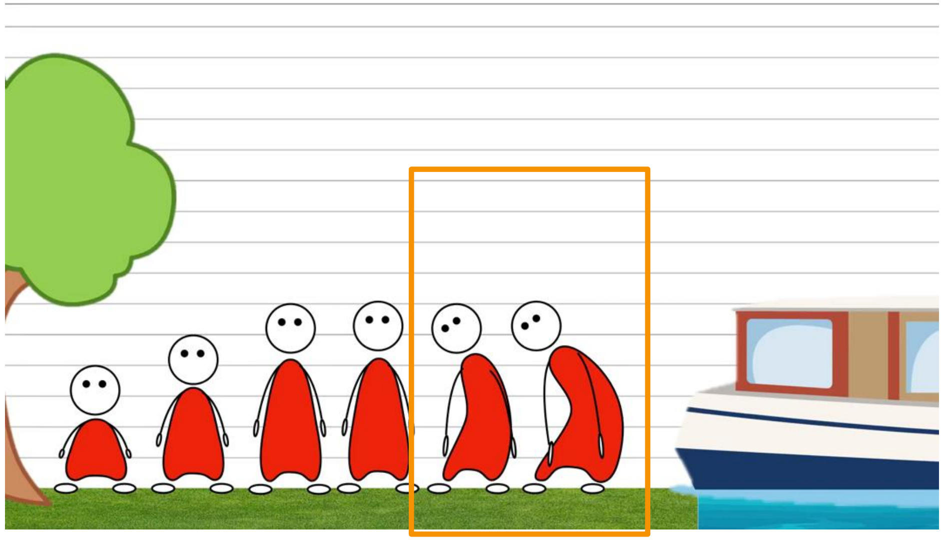
~~alternative explanation: perceptual salience~~

~~adults and children ignored sampling processes, and~~ **~~generalized directly from the sample,~~**
~~weighting the taller Zarpies more in the skewed condition,~~ ~~since scrunching makes them salient~~

not skewed



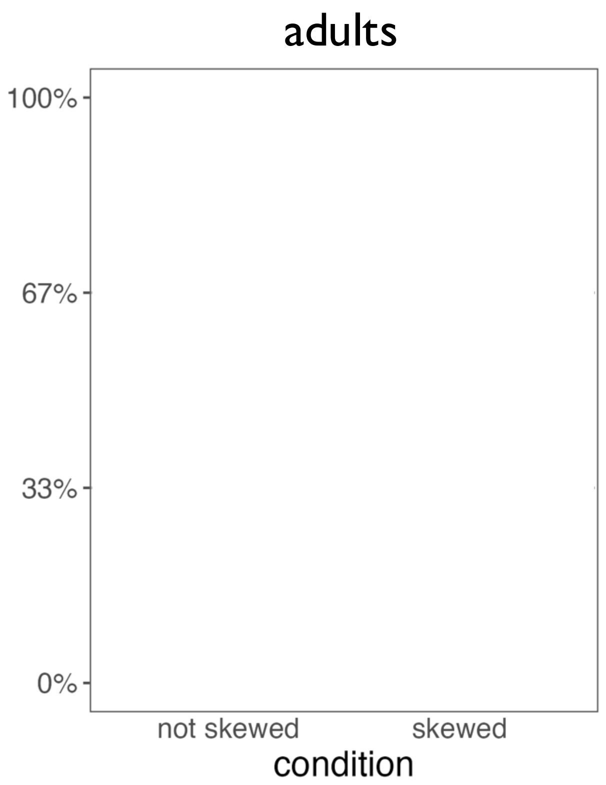
skewed

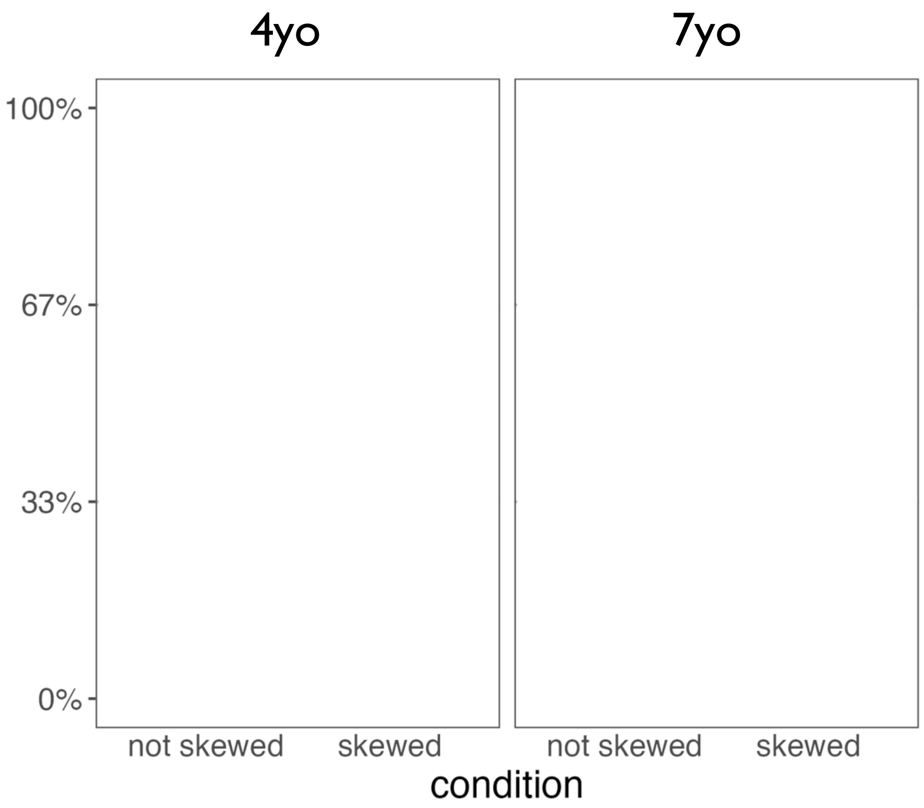


sample observation scene matched
in both conditions, 2 Zarpies scrunch and boat is short

Study 2

If adults' and children's success is not due to perceptual salience,
they should continue to think the social group was taller in the skewed condition,
even when scrunching is matched.





adults

Zarpies on Zarpie island are
- shorter
- the same
- taller

than the Zarpies who visited

4yo, 7yo

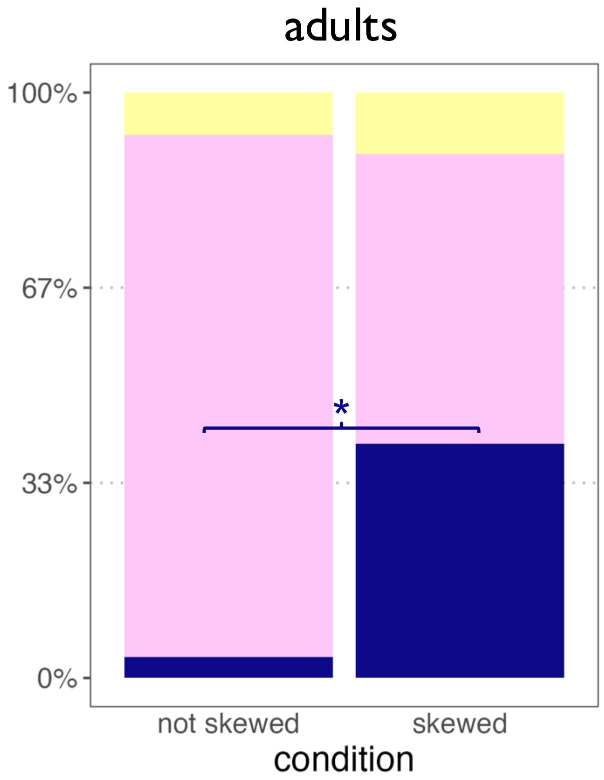
Who's taller?
- Zarpies who visited
- the same
- Zarpies on Zarpie island

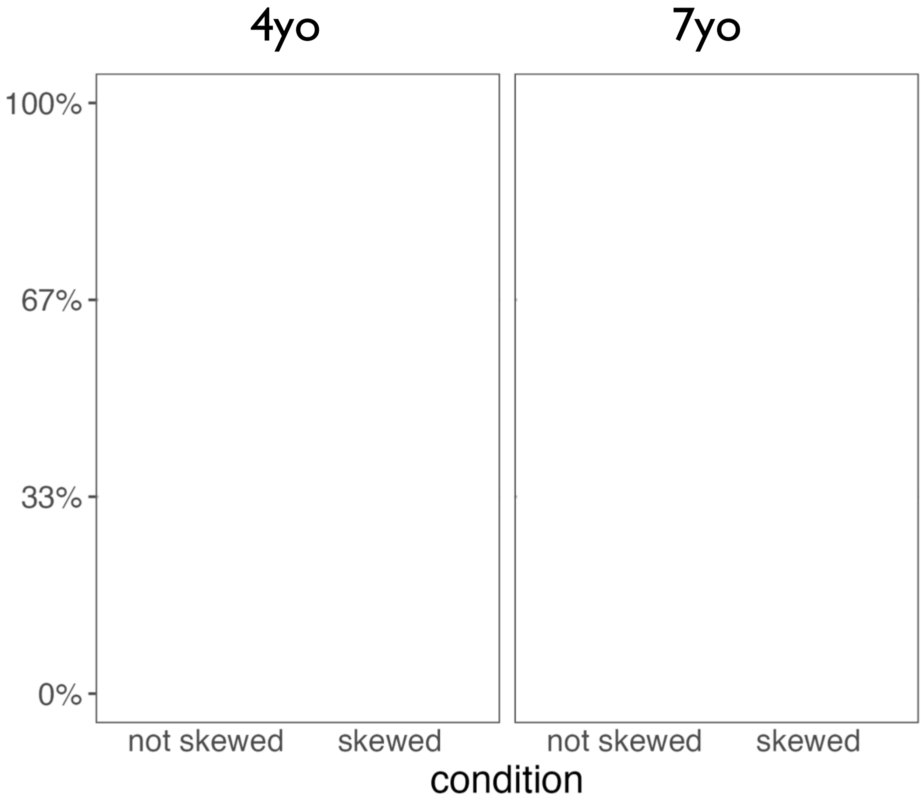
2a: **adults** (n=200) on Prolific
2b in progress: **4yo** (n=271/295), **7yo** (n=293/295) on PANDA

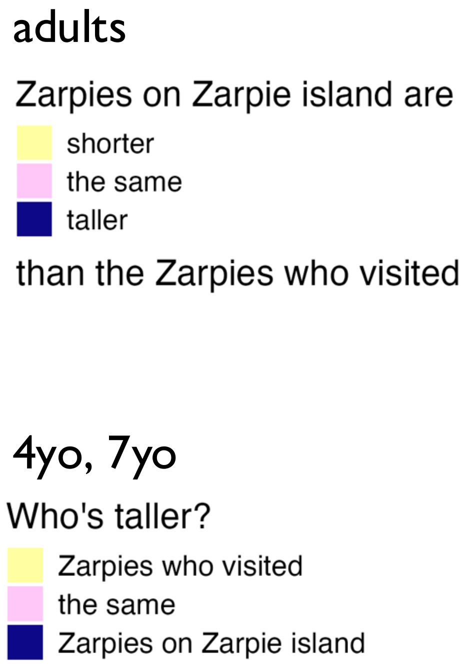
Study 2

Adults' success in Study 1 cannot be attributed to perceptual salience.

Adults were more likely to think the social group was taller when the sample was skewed towards shorter individuals, even when scrunching is matched.







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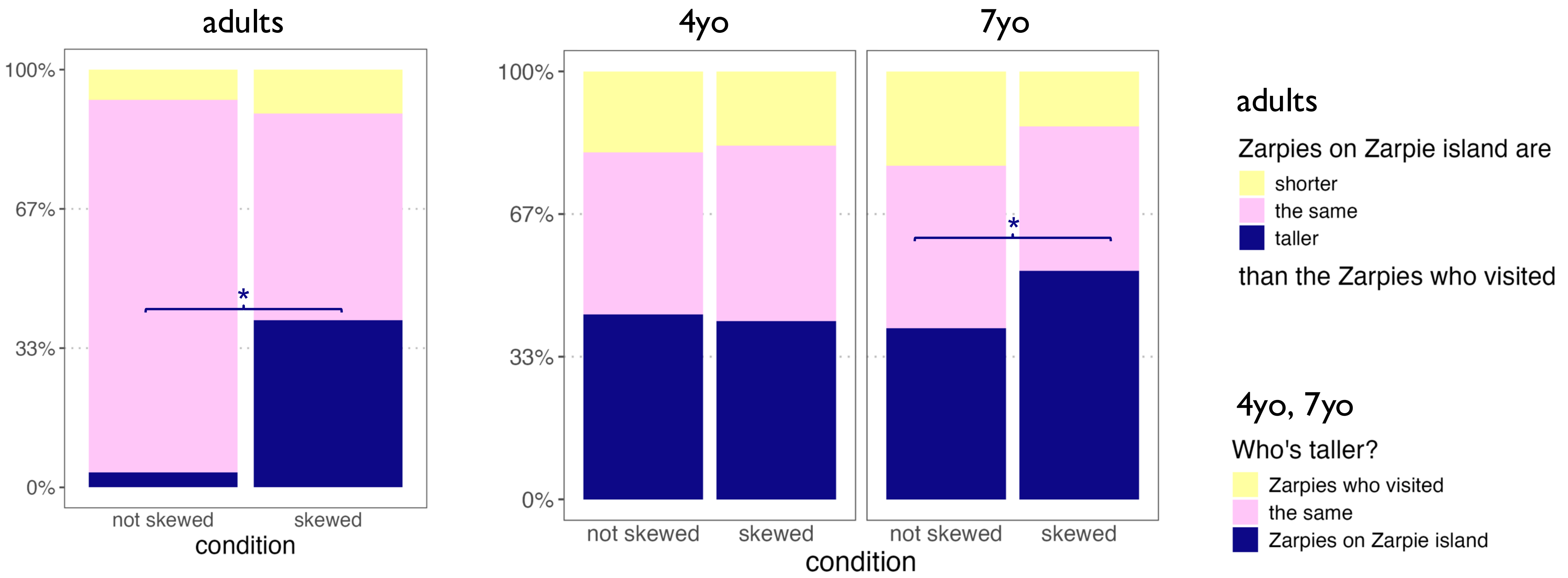
2b in progress: **4yo** (n=271/295), **7yo** (n=293/295) on PANDA

adults:
"Zarpies on Zarpie island" ~ condition: z=4.61, p<.001

Study 2

Adults & 7yo's success in Study 1 cannot be attributed to perceptual salience.
However, 4yo fail to adjust their inferences, suggesting 4yo struggle to adjust their inferences about social groups in response to skewed samples.



2a: **adults** (n=200) on Prolific
2b in progress: **4yo** (n=276/295), **7yo** (n=293/295) on PANDA

4yo:
"Zarpies on Zarpie island": *LR* $\chi 2$=0.03, *p*=.86, BF=0.24

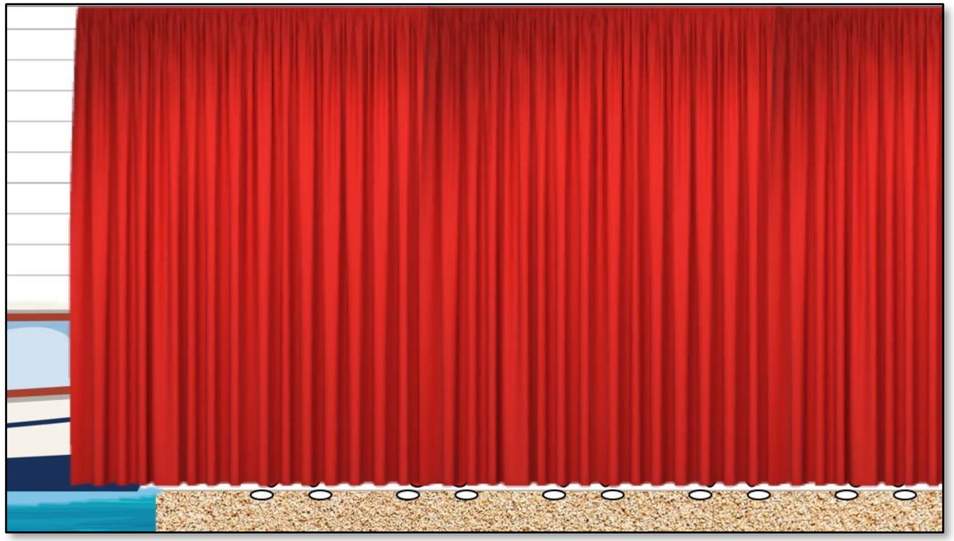
7yo:
"Zarpies on Zarpie island": *LR* $\chi 2$=5.28, *p*=.022; BF=2.84

thanks to Susan Carey for pointing out this concern!

Study 2

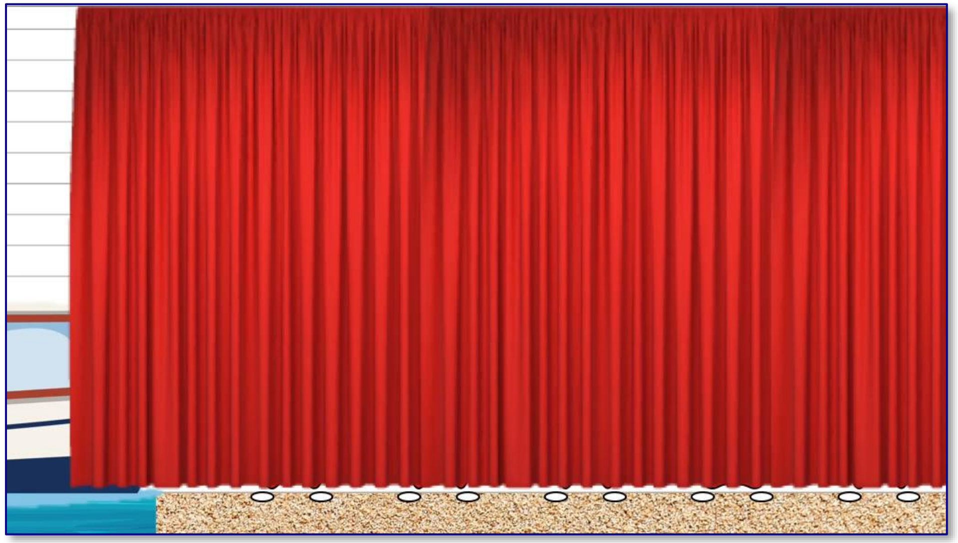
alternative explanation: revealed population information
adults' and 7yo's success in Studies 1-2 may come from **reasoning about revealed population information during the boarding scene**, ignoring the sample and sampling process

not skewed



successful boarding suggests
the Zarpies behind curtain are shorter than boat

skewed

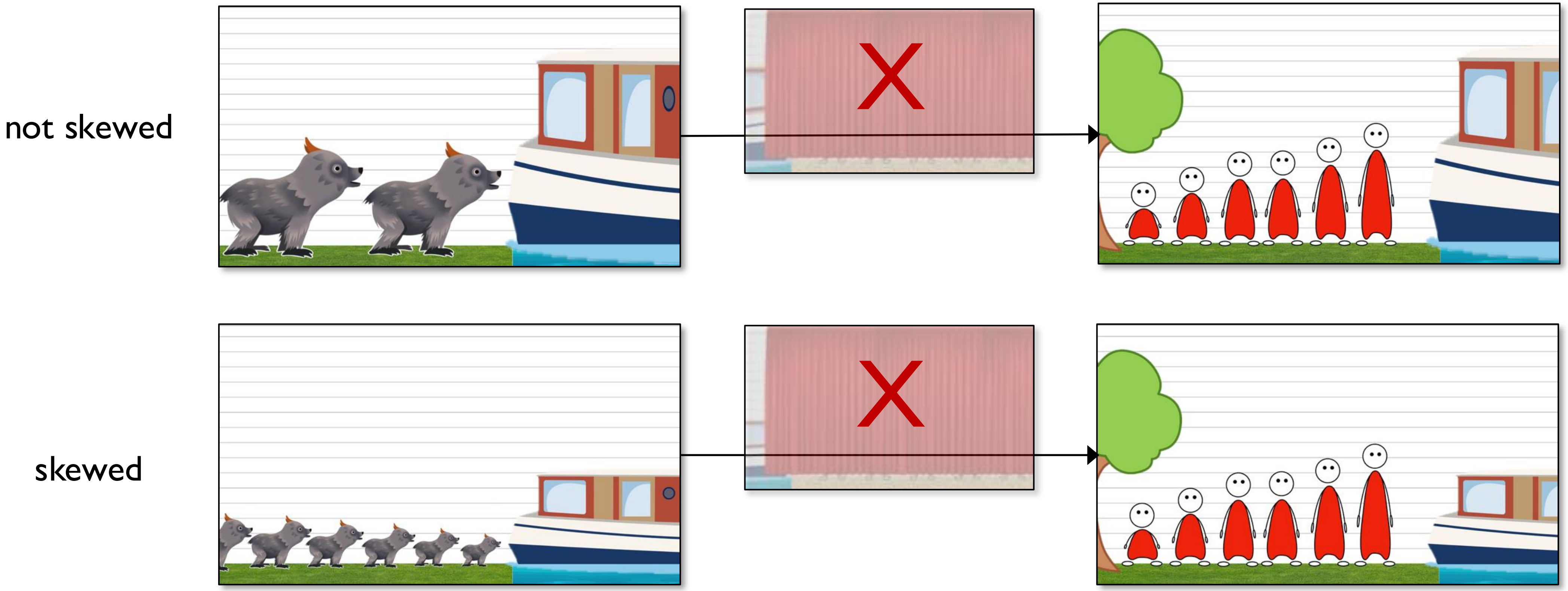


failure of most attempts to board suggests
most Zarpies behind curtain are taller than boat

Study 3

addressing revealed population information

warmup | training | ~~sampling process~~ | sample observation | inference

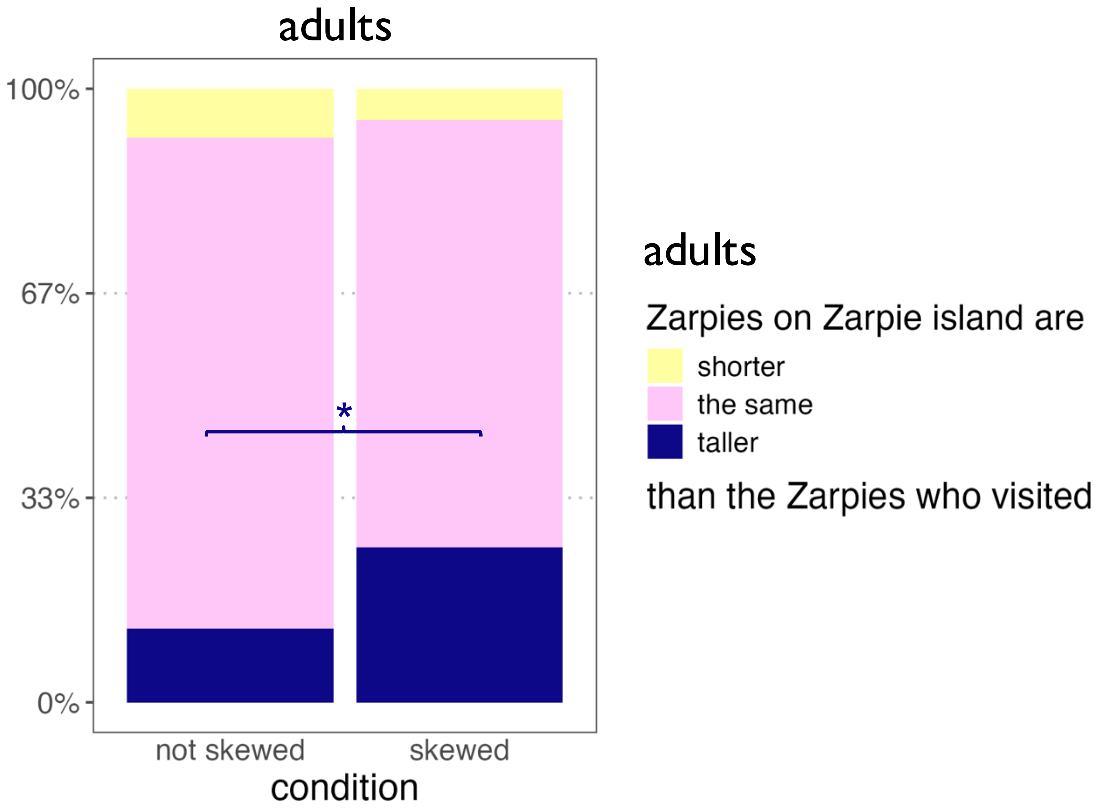


3: adults (n=100/condition) on Prolific

Study 3

addressing revealed population information

Adults' success in Studies 1-2 cannot be explained by revealed population information, suggesting adults are indeed capable of adjusting their beliefs.



However, the task remains difficult, even for adults.

3: adults (n=100/condition) on Prolific

People can adjust their generalizations about social groups from skewed samples, but it is challenging and may not available before middle childhood.

Children and adults' inferences about social groups in response to sampling skew

- Adults account for skewed samples of social groups and **adjust their inferences** about the groups accordingly.
- 7yo may be able to as well, while 4yo do not.
    - Contrast with infants' ability to suspend inference from samples skewed by an agent's preference! Perhaps the difficulty lies in:
        - the ability to **adjust** (vs suspend) **inferences**, and/or
        - the ability to reason about **structural** (vs agentic) **sources of skew**
        - Next study: do children **adjust** inferences about social groups from samples skewed by **agent preference**?

Children and adults' inferences about social groups in response to sampling skew

Implications for stereotypes

- Young children may be **unable to fend off the effects of exposure to skewed social samples**, which could lead to **stereotype development.**
- However, the developing ability to adjust inferences to account for skewed social samples could suggest **an intervention against stereotypes** in older children and adults:
  - **fostering an understanding of the social processes** that shape who we see (and who we don't see).

Children and adults' inferences about social groups in response to sampling skew

Related questions

- Given an existing belief about a social group, and a sample that looks different, can people **infer structural skew** must have occurred?
  (c.f. social structures are typically difficult to notice/attribute effects to)
- Beyond social groups, could same cognitive machinery allow people to form better beliefs from **limited and skewed information**?
  (forming beliefs in constrained epistemic environments,
  e.g., under conditions of censorship)

Thanks!

Mark Ho

Marjorie Rhodes

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Neela Rao

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NYU Postdoctoral Trailblazer Award
AWIS Bridge Grant Award
NSF SBE Postdoctoral Research Fellowship (formerly)

CDSC

Computation and Decision-Making Lab

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